

TrES-3b

R-2819

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_130810 · created 2026-09-05T02:17:57+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2456514.8429 +/- 0.0011 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.135 +/- 0.054
Transit depth (Rp/Rs)^2	1.83 +/- 1.45 [%]
Orbital Inclination (inc)	80.86 +/- 2.71
Airmass coefficient 1 (a1)	0.98 +/- 0.022
Airmass coefficient 2 (a2)	0.0143 +/- 0.0067
Scatter in the residuals of the lightcurve fit is	2.22 %
Best Comparison Star	#3 - [118, 147]
Optimal Aperture	3.97
Optimal Annulus	10.65
Transit Duration (day)	0.035 +/- 0.028

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.135 $\pm$ 0.054 vs 0.16309 (deviation 0.51 σ)
Duration fit vs published	0.035 d vs 0.0567 d (deviation 0.77 σ)
Mid-time O-C	1.6 min
Depth SNR	2.5
Residual scatter	2.22 %

Light curve

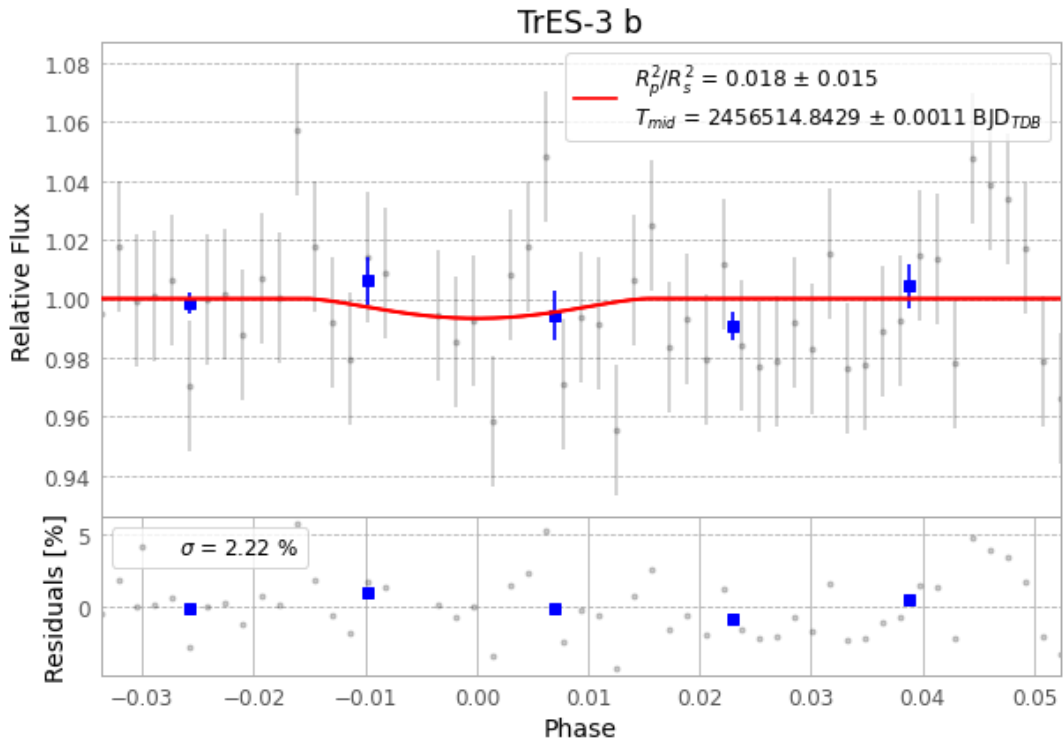


Figure 1 — FinalLightCurve_TrES-3 b_2013-08-10.png (SHA-256 378ac02320a64f42...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [190, 304], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 574d5e0f0eccfdd2...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

55 FITS frames (36 MB), TRES-3130810070613.FITS → TRES-3130810094812.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-130810.log (86677dee5414...), FinalLightCurve_TrES-3 b_2013-08-10.png (378ac02320a6...), FinalLightCurve_TrES-3 b_2013-08-10.csv (4006678d724b...)

Compute resources

Wall time 268.5 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-09-05 02:18Z.